

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

df = pd.read_csv("Sleep_health_and_lifestyle_dataset.csv")

print("Dataset shape:", df.shape)
print("\nColumns:")
print(df.columns)
```

Dataset shape: (374, 13)

Columns:

```
Index(['Person ID', 'Gender', 'Age', 'Occupation', 'Sleep Duration',
      'Quality of Sleep', 'Physical Activity Level', 'Stress Level',
      'BMI Category', 'Blood Pressure', 'Heart Rate', 'Daily Steps',
      'Sleep Disorder'],
      dtype='object')
```

Calculation of Sleep Score

```
quality_score = (df["Quality of Sleep"] / 10) * 100

duration_score = 100 - abs(df["Sleep Duration"] - 8) * 25
duration_score = duration_score.clip(0, 100)

activity_score = (df["Physical Activity Level"] / 90) * 100
activity_score = activity_score.clip(0, 100)

disorder_score = np.where(
    df["Sleep Disorder"].isna(), 100, 0
)

df["Sleep Score"] = (
    0.45 * quality_score +
    0.30 * duration_score +
    0.15 * activity_score +
    0.10 * disorder_score
)
```

Calculation of Mental score and Stress score

```
df["Stress Score"] = (
    (10 - df["Stress Level"]) / 9
```

```

    ) * 100

df["Mental Health Score"] = (
    0.60 * df["Sleep Score"] +
    0.40 * df["Stress Score"]
)

```

Differenr Mental Health Categories

```

df["Mental Health Category"] = pd.cut(
    df["Mental Health Score"],
    bins=[-1, 50, 70, 100],
    labels=["Poor", "Moderate", "Good"]
)

print("\nSample results:")
print(
    df[
        [
            "Sleep Duration",
            "Quality of Sleep",
            "Physical Activity Level",
            "Stress Level",
            "Sleep Score",
            "Mental Health Score",
            "Mental Health Category"
        ]
    ].head(10)
)

```

Sample results:

	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level
0	6.1	6	42	
1	6.2	6	60	
2	6.2	6	60	
3	5.9	4	30	
4	5.9	4	30	
5	5.9	4	30	
6	6.3	6	40	
7	7.8	7	75	
8	7.8	7	75	
9	7.8	7	75	

	Sleep Score	Mental Health Score	Mental Health Category
0	59.750000	53.627778	Moderate
1	63.500000	46.988889	Poor
2	63.500000	46.988889	Poor
3	37.250000	31.238889	Poor
4	37.250000	31.238889	Poor
5	37.250000	31.238889	Poor
6	50.916667	43.883333	Poor
7	82.500000	67.277778	Moderate
8	82.500000	67.277778	Moderate
9	82.500000	67.277778	Moderate

```

X = df[["Sleep Score", "Stress Level"]]
y = df["Mental Health Category"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

model = RandomForestClassifier(
    n_estimators=200,
    max_depth=5,
    random_state=42
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

```

Model evaluation

```

accuracy = accuracy_score(y_test, y_pred)

print("\n-----")
print("Model Performance")
print("-----")

print("Accuracy:", round(accuracy * 100, 2), "%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

```

```

-----
Model Performance
-----
Accuracy: 98.67 %

Classification Report:

```

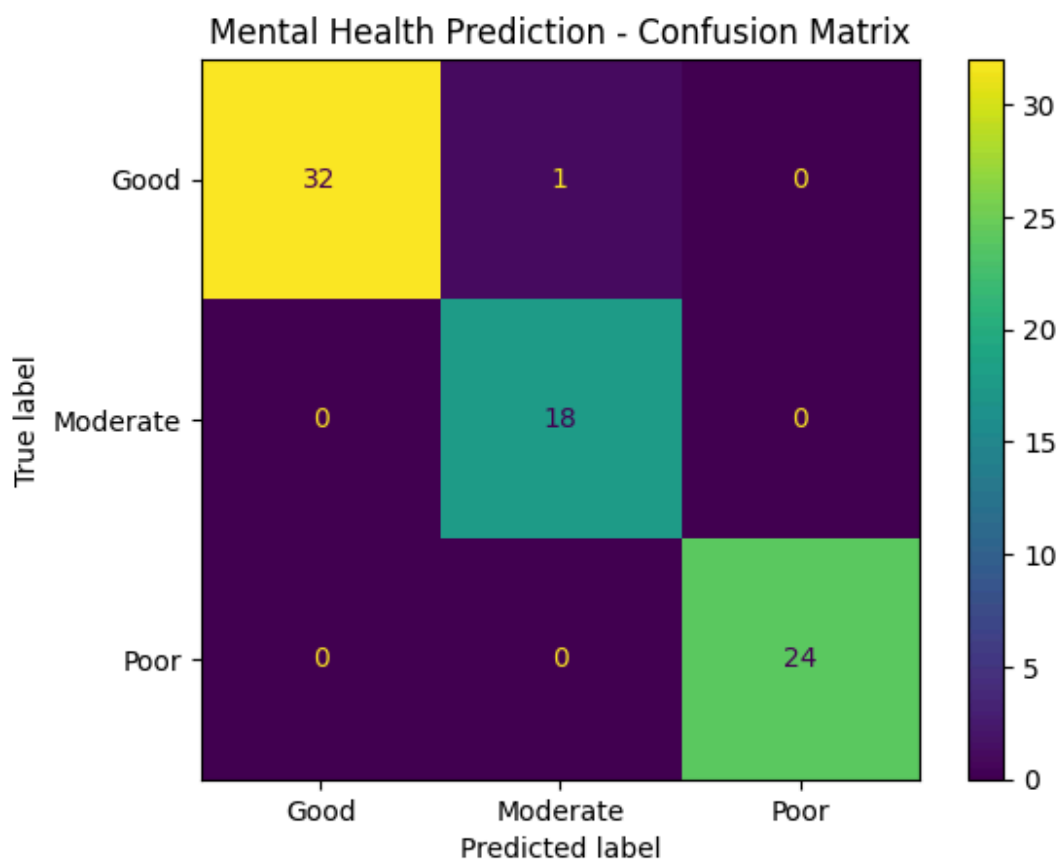
	precision	recall	f1-score	support
Good	1.00	0.97	0.98	33
Moderate	0.95	1.00	0.97	18
Poor	1.00	1.00	1.00	24
accuracy			0.99	75
macro avg	0.98	0.99	0.99	75
weighted avg	0.99	0.99	0.99	75

Confusion Matrix

```
cm = confusion_matrix(y_test, y_pred)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=model.classes_
)

disp.plot()
plt.title("Mental Health Prediction - Confusion Matrix")
plt.show()
```

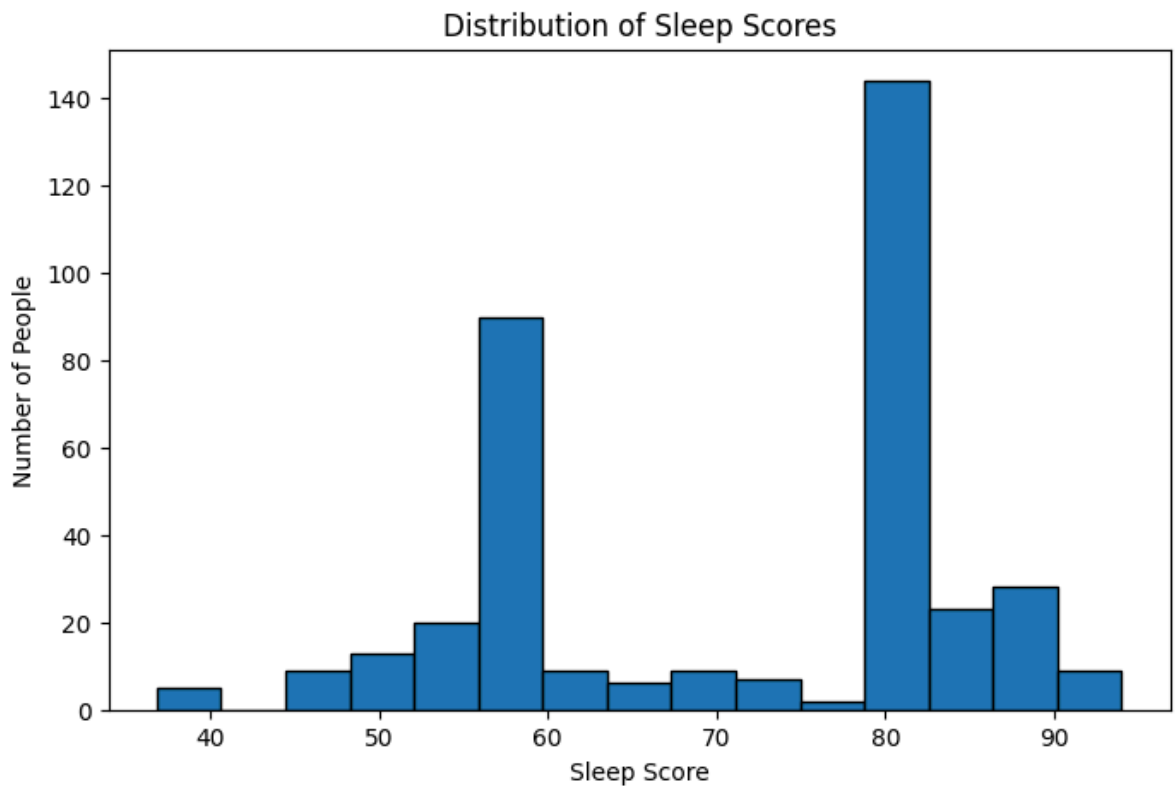


Sleep Score Distribution

```
plt.figure(figsize=(8, 5))

plt.hist(
    df["Sleep Score"],
    bins=15,
    edgecolor="black"
)

plt.xlabel("Sleep Score")
plt.ylabel("Number of People")
plt.title("Distribution of Sleep Scores")
plt.show()
```

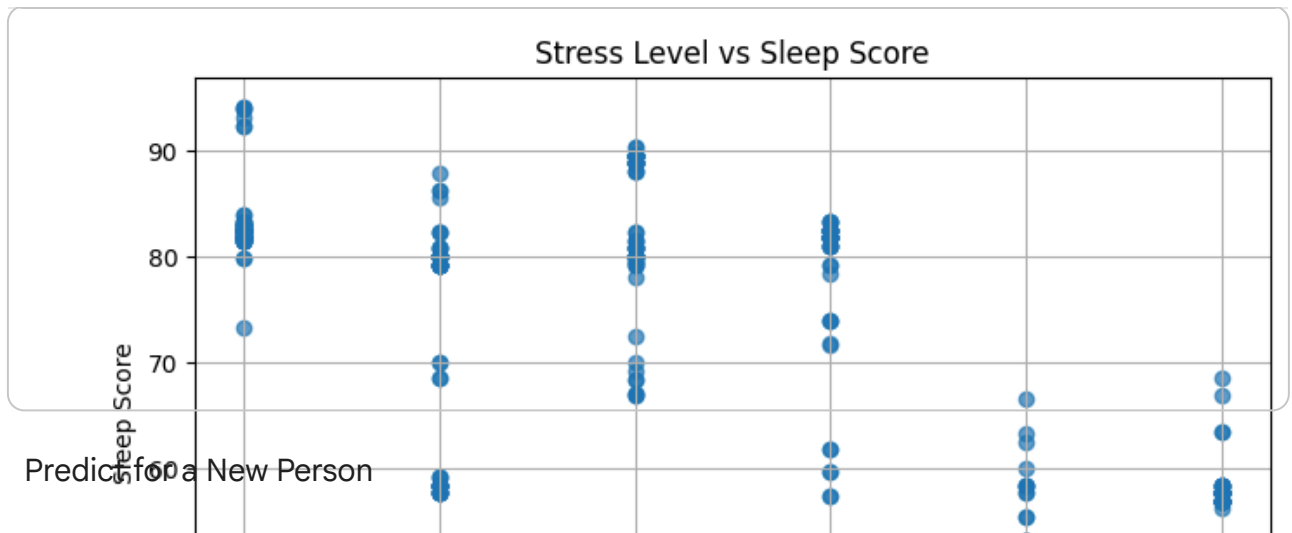


Stress Level vs Sleep Score

```
plt.figure(figsize=(8, 5))

plt.scatter(
    df["Stress Level"],
    df["Sleep Score"],
    alpha=0.7
)

plt.xlabel("Stress Level")
plt.ylabel("Sleep Score")
plt.title("Stress Level vs Sleep Score")
plt.grid(True)
plt.show()
```



```
new_person = pd.DataFrame({
    "Sleep Score": [75],
    "Stress Level": [4]
})

prediction = model.predict(new_person)

print("\nNew Person Prediction:")
print("Sleep Score:", new_person["Sleep Score"].iloc[0])
print("Stress Level:", new_person["Stress Level"].iloc[0])
print("Predicted Mental Health:", prediction[0])
```

```
New Person Prediction:
Sleep Score: 75
Stress Level: 4
Predicted Mental Health: Moderate
```